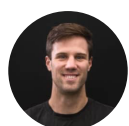


June 17, 2024

Reliability and validity of the Metric VBT beta

It is early days for our VBT technology, but we can confidently say Metric works very well already! An internal study into the accuracy of the Metric beta



Jacob Tober
Co-founder

Note: This data was collected in October 2021 using Metric v0.3 prior to public release. Since then the app has also been [externally validated](#)



and numerous tracking improvements have been released into the current publicly available version of the app.

[Download Metric on iOS here →](#)

How accurate is Metric VBT?

Many of our beta testers have been asking us about how Metric stacks up against alternative VBT measurement devices. If you don't currently have access to a VBT measurement device, how can you know if you should rely on Metric?

It is very early days for our technology, but the short answer right now is: *Metric performs incredibly well!*

To give a longer answer, I want to share with you the results of an internal study we have conducted comparing Metric (Version 0.3, March 2022) with a high end MOCAP (Motion Capture) system. The below article is a summary of the full report. (You can download the [PDF of the full study here](#) including the raw data for all repetitions.)

We are determined to build a tool that is reliable and valid, making it possible for sports scientists, coaches and athletes to improve their performance by leveraging velocity monitoring. To do that we had to throw out old thinking. Today, I am more sure than ever that the future of VBT measurement is computer vision.

In other blog posts and social media comments Thomas and myself have mentioned that we are developing Metric against MOCAP rather

than existing VBT hardware. We have several VBT devices in our gym from the main manufacturers, and we know from experience that they all return valid but divergent results! Despite what some may claim, MOCAP is the real “gold standard” of measurement. Unfortunately a MOCAP system is not at all portable, and not at all affordable!

Whether using a string, lasers, accelerometer or video, methods of measuring VBT on the market today are all making an indirect assessment of the movement of the barbell. They all use algorithms to smooth their data and infer the barbell velocity, and they all use algorithms to determine rep start and stop times, bar path and so-on. Anyone claiming otherwise is not acting in good faith.

Details of internal study

Research lead: Jacob Tober

Team: David Tober, Mason Lauder, Durham McInnis, Thomas McInnis

*Location: TrackLab motion capture studio, Yarraville, Victoria, Australia
3013*

Date of data collection: 28th of October 2021

*Study purpose: **Assess the validity and reliability of the MetricVBT app to track typical barbell exercises at fast and slow movement velocities.***

Introduction

Despite the growing support and interest in the use of velocity in the weight room there remains a generally low rate of adoption for VBT

across the wider resistance training community. Cost (either financial or time), lack of knowledge, informational overwhelm, and user experience issues when implementing technology leaves many unable to incorporate velocity into their training despite a desire to do so.

There remains a need for simple, affordable solutions to tracking velocity in the gym. Utilising only a smartphone device and the inbuilt camera system, MetricVBT works by way of a sophisticated computer vision algorithm specifically designed to recognise, track and calculate the displacement and velocity of barbell movements with regulation weight plates.

Method

- I acted as the participant of this study.
- Video was captured using a smartphone (iPhone XS, Apple*,* California) using the in-built camera application. Recording was completed in a vertical orientation at 1080p resolution with the 60 frames per second setting enabled. The smartphone device was positioned approximately 2.5 metres directly from the end of the barbell with the device positioned in a tripod at waist height.
- Raw positional data was simultaneously captured using the Tracklab MOCAP system (24 x *OptiTrack* cameras, and accompanying software, *Motive 3* - NaturalPoint Inc., Oregon) using a sampling rate of 120Hz. A single marker was positioned in the exact centre point of the barbell end.

- I completed two sets of eight repetitions for the deadlift, front squat, and bench press, completing a total of 48 repetitions. Each exercise was completed for a controlled, slow set (TEMPO) and a maximum effort, fast set (MAX). All sets were completed with 40kg – approximately 30–40% of my 1RM.

Two sets each of Deadlifts, Front Squat, and Bench Press were performed.

- The video footage from the smartphone and the MOCAP positional data was then analysed using the MetricVBT (Beta v0.3) algorithm to count repetitions (reps), and calculate mean concentric velocity (VEL) and concentric range of motion (ROM) for every rep.
- Overall validity for both VEL and ROM was calculated comparing the Smartphone device footage with the MOCAP data as baseline. Reliability was assessed using absolute differentials and percentage variation from the MOCAP baseline. Correlation values for VEL and ROM were also established.

Results summary

- MetricVBT was able to accurately detect 48/48 repetitions from the smartphone footage with no additional “phantom” reps recorded or repetitions missed.
- The smartphone footage and MOCAP data produced strong correlations across all recorded reps for both ROM ($R = 0.9862$), and VEL ($R = 0.9841$).

- Average absolute difference was 0.026m/s (3.3%) for VEL, and 2.49cm (4.0%) for ROM. Individual rep differentials from MetricVBT to the MOCAP baseline ranged from -0.00m/s to +0.08m/s (0.0% to 7.2%) for VEL and 0.04cm to 5.6cm (0.1% to 7.8%) for ROM.

ROM and VEL correlation for all repetitions. Line represents a perfect (1.0) correlation.

VEL and ROM results for each individual repetition for MetricVBT and MOCAP.

Discussion

Our results show that MetricVBT v0.3 is able to accurately detect repetitions and analyse concentric mean velocity and range of motion for standard strength exercises.

We understand that to be useful, a measurement device needs to be both valid and reliable. The user must get data that is relevant to the real-world, and get it repeatedly.

Metric v0.3 has shown it is able to achieve both these objectives with a tight correlation to the results produced by the MOCAP system across a range of velocities and exercises.

While Metric did under-report both VEL and ROM on almost all repetitions, it was able to do this within a tight range. This means users can be confident results from MetricVBT is reflective of a change in performance and not due to measurement error.

Even though Metric is still in Beta, we strongly believe that it's simplicity in use and negligible implementation cost make it a compelling option for coaches and athletes at all levels looking to incorporate velocity into their training.

It is also worth stating that the Metric algorithm is less than a year old! We are continuously updating Metric as we invest in further research and development. The pace of improvement in such a short time is stunning, and each update brings greater reliability and accuracy, plus further accomodation of edge cases and more and more exercises. We can't wait to continue developing Metric's features [as we revolutionise how VBT is utilised](#).

It is inevitable that velocity based training using computer vision will be the obvious choice for most practitioners in the future. Computer vision apps are easy to use, cost effective, and portable.

We plan for Metric to launch on the App Store for iPhone later this month, but in the mean time the beta program is still open for those looking for an early preview.

UPDATE: Metric is now available on iOS and can be [downloaded for free at this link](#).

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By lifters for lifters

Metric was built with lifters and coaches front of mind.

Focused analytics and powerful workout tracking tools for Powerlifting, Strength and conditioning, CrossFit, Weightlifting, and anyone serious about their strength training.



METRIC

Barbell velocity tracking

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